Research Article Vol. 33, No. 18 / 25 Aug 2025 / Optics Express 36646

Optics EXPRESS

Accelerated Φ-OTDR Event Recognition via Tabular Few-Shot ML Techniques

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Abstract: Distributed optical fiber sensing systems based on phase-sensitive optical timedomain reflectometry (Φ-OTDR) play a crucial role in infrastructure monitoring. However, event classification in these systems is hindered by limited data availability, high computational complexity, and slow model training. In this paper, we present an event classification method that integrates tabular machine learning with few-shot learning (FSL) through a pre-trained tabular prior-data fitted network (TabPFN) model. This method transforms the comprehensive set of time-domain statistical features (e.g., mean, peak-to-peak) and frequency-domain energy distribution features extracted from a conventional Φ-OTDR into structured tabular data, enabling subsequent machine learning processing. Specifically, the proposed scheme achieves a five-fold cross-validation accuracy of 90.0% ± 6.24% and an independent test accuracy of 89.0% under extremely small sample conditions and with only 10 samples per class available. Furthermore, with an increasing sample size, the performance further improves, reaching 97.78% ± 1.88% (at 60 samples per class) in cross-validation and 100% (at 50 samples per class) in independent testing, while fitting consistently completes within seconds. Comparative analysis against both strong traditional machine learning models and representative meta-learning FSL baselines further validates the superiority of our approach in few-shot settings. These impressive results and low computational overhead during application pave the way for rapid industrial deployment of Φ-OTDR systems in applications such as oil and gas pipeline monitoring.

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1. Introduction

With the rapid urbanization and industrialization, the construction and maintenance of modern infrastructure have become increasingly complex and critical. Among various monitoring technologies, the phase-sensitive optical time domain reflectometer (Φ-OTDR) stands as a pivotal technology in distributed fiber optic sensing systems owing to its remarkable sensitivity and long-distance monitoring capabilities [1]. This technology is widely employed in the structural health monitoring of critical infrastructures, including construction sites, transportation infrastructure, transmission pipelines, and submarine power cable monitoring [2–7].

Meanwhile, machine learning techniques have demonstrated robust classification capabilities across diverse fields. The Φ-OTDR system is increasingly applied to identify event types occurring in the vicinity of optical fibers. To handle the complexities of Φ-OTDR signals and prepare them for analysis, a range of preprocessing and feature engineering strategies are utilized. These encompass noise reduction techniques [8], time-frequency representation with STFT [9], signal component separation via VMD [10], and feature extraction methods like MFCC [11] and Markov transition fields (MTF) [12]. Subsequent event classification efforts have utilized both traditional machine learning algorithms, such as SVM [13] and RF [14], and deep

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learning approaches like CNN [13] and ViT [15]. To address data scarcity, methods like CycleGAN [16]

#571235 https://doi.org/10.1364/OE.571235
Journal © 2025 Received 19 Jun 2025; revised 27 Jul 2025; accepted 6 Aug 2025; published 19 Aug 2025

and transfer learning [17] have been explored. Furthermore, dedicated meta-learning approaches, such as those based on Model-Agnostic Meta-Learning (MAML) [18] and Prototypical Networks [19], have recently been investigated to specifically tackle the few-shot challenge in Φ-OTDR. However, CycleGAN necessitates computationally expensive adversarial training while transfer learning critically depends on the similarity between source and target domains. Other paradigms like zero-shot learning (ZSL), aiming to classify entirely unseen event types in Φ-OTDR systems [20], also address data limitations but differ in problem formulation from our few-shot learning focus. In contrast, our TabPFN-based approach leverages prior knowledge through pre-training, eliminating the need for domain alignment or complex adversarial optimization.

Current mainstream research focuses on reducing the required data volume while maintaining the accuracy and efficiency of event classification. This strategy aims to lower data acquisition costs and facilitate the application of Φ-OTDR event classification in industries where data collection is challenging.

Few-shot learning (FSL) [21] offers a promising solution to limited data, enabling rapid adaptation to new categories with few labeled examples, as demonstrated in fields like image classification. FSL techniques effectively extract key information from scarce data, allowing models to classify novel events even under data-scarce conditions. For Φ-OTDR intrusion detection, an FSL-based approach can reduce reliance on large datasets, enhance recognition of low-frequency events, and improve overall system adaptability and sensitivity while maintaining high accuracy.

While conventional deep learning in areas like image classification often demands extensive data, few-shot learning paradigms (including one-shot learning (OSL) and ZSL) [22–24] have emerged to enable effective learning from minimal samples.

Particularly for one-dimensional signals like those in Φ-OTDR systems, where conventional feature representation can be challenging with limited data, FSL (as demonstrated in fields like EEG signal analysis [25]) offers significant advantages by reducing dataset dependency and training overhead. This work explores this potential to enhance Φ-OTDR event recognition in data-scarce scenarios.

This paper proposes a Φ-OTDR event classification method based on TabPFN [26] for event type identification, aimed at intrusion detection in perimeter security. TabPFN utilizes a pretrained Transformer architecture to perform direct inference without traditional gradient updates. This significantly enhances the model's generalization ability in low-data scenarios, rendering it highly suitable for environments with limited data. This represents the first application of TabPFN in a Φ-OTDR system. The proposed method extracts features from Φ-OTDR signals and converts them into tabular form, enabling efficient and accurate intrusion signal recognition even with a small number of samples. Due to its inherently low computational overhead and rapid task adaptation capability, TabPFN enables the Φ-OTDR to perform event recognition efficiently even in data-scarce or resource-constrained environments, meeting the practical demands of perimeter security. The main steps are delineated as follows:

1. Preprocess the raw Φ-OTDR data and extract essential features to construct tabular representations;

Optics EXPRESS

2. Apply the pre-trained TabPFN network to the Φ-OTDR event classification task, utilizing its rapid In-Context Learning based fitting process for few-shot learning (FSL) and tabular machine learning (TML) on tabularized signal data, reducing raw data requirements;

3. Validate the method's superiority through a rigorous comparative analysis on a public dataset, benchmarking against both strong traditional machine learning models and representative meta-learning FSL baselines.

This work's primary contributions are: applying TabPFN to few-shot Φ-OTDR event classification, systematically evaluating its capabilities in extreme data scarcity, and a rigorous comparison with traditional machine learning baselines, assessing its relative efficacy and efficiency.

2. Tabular machine learning and few-shot learning

2.1. Few-shot learning

Few-Shot Learning (FSL) is a paradigm that enables rapid adaptation using a small number of samples, enabling effective network training despite limited data. This approach reduces the need for large amounts of labeled data and circumvents the necessity of training from scratch, making it particularly suited for accelerating the fitting process of the Φ-OTDR event recognition network.

The fundamental structure of the Φ-OTDR system used in this study is depicted in Fig. 1. This real-time optical fiber sensing system operates by injecting probe light pulses into the sensing fiber and capturing the backscattered Rayleigh traces. These traces, organized temporally, form a spatiotemporal data matrix where rows represent distance and columns represent time. External vibrations along the fiber induce localized intensity fluctuations in the Rayleigh backscatter, which are detected and localized. Subsequently, the spatiotemporal data segment corresponding to a located vibration is extracted as an event sample for feature engineering and tabular machine learning classification.

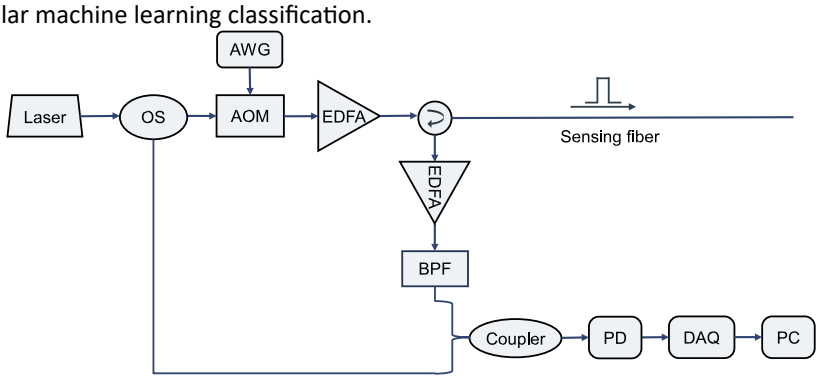

Fig. 1. Structural diagram of the Φ-OTDR system: OS: Optical Splitter, AOM: Acousto-Optic Modulator, AWG: Arbitrary Waveform Generator, EDFA: Erbium-Doped Fiber Amplifier, BPF: Bandpass Filter, PD: Photodetector, DAQ: Data Acquisition Card, PC: Personal Computer

2.2. Tabular machine learning and pre-trained model TabPFN

Tabular machine learning is designed to model and predict structured tabular data, such as databases or spreadsheets. It leverages the features (columns) of each row to train predictive models. Unlike unstructured data (e.g., text or images), tabular data is characterized by its heterogeneity, where identical values may represent distinct entities or concepts across different contexts, in contrast to the consistent semantics of words in language modeling.

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This research utilizes TabPFN, a pre-trained Transformer model specifically designed for tabular data, which uniquely leverages the power of in-context learning (ICL) [27]. In its application, ICL enables TabPFN to make predictions for new, unseen samples by conditioning on a few provided input-output examples (the "context") from the target task, without requiring retraining or fine-tuning of its underlying pre-trained parameters. The model's powerful capability is attributed to its origin: we use the publicly available weights [26], which were obtained from a domain-agnostic pre-training on a massive corpus of approximately 130 million synthetic tabular datasets. Consequently, no correlation with the specific Φ-OTDR task in this study is required. The model itself consists of a 12-layer Transformer encoder, with its operational scope defined by a 1000-point/100-feature context window. Rather than emulating specific Φ-OTDR signal characteristics in the synthetic data, the underlying principle is to train the model with a general, robust Bayesian-like reasoning ability. This capability is then applicable to any tabular data, including feature tables derived from Φ-OTDR signals. This domain-agnostic nature means the only format requirement is that the task data be structured as a feature table, making the approach highly flexible. The potential of ICL is underscored by recent studies demonstrating Transformers' acquisition of algorithmic capabilities directly from context [28–31], extending earlier work on Prior-data Fitting Networks (PFNs) [32]. ICL thus significantly expands the horizon for applying complex models, particularly where analytical solutions are scarce.

TabPFN operates by generating a large corpus of synthetic tabular datasets and training a Transformer to solve corresponding prediction tasks. Unlike traditional methods requiring manual solutions for challenges like missing values, TabPFN autonomously learns strategies by addressing synthetic tasks that embed such issues. It uses ICL as a framework for exemplar-based declarative algorithm programming, shifting algorithm design from explicit instructions to defining input-output examples. This approach is applied here to tabular machine learning, yielding a powerful prediction algorithm.

Built on the Transformer architecture, TabPFN employs self-attention to process tabular data. The attention mechanism is defined as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where Q, K, and V represent the query, key, and value matrices, respectively, and d_k is the dimension of the key vectors. TabPFN adapts this through a two-way attention mechanism: each cell attends first to other features within its row (intra-sample) and then to the same feature across rows (inter-sample), capturing both sample-specific and feature-specific relationships.

Through ICL, TabPFN predicts test sample labels based on training sample context, approximating the Bayesian posterior predictive distribution:

$$p(y_{test}|X_{test}, X_{train}, y_{train}) = \int p(y_{test}|X_{test}, \theta)p(\theta|X_{train}, y_{train})d\theta \tag{2}$$

where θ denotes model parameters. Pre-training on diverse synthetic datasets enables TabPFN to approximate this distribution effectively, ensuring robust performance across tabular tasks.

For classification, TabPFN outputs a probability distribution over classes, trained by minimizing the cross-entropy loss:

$$L = -\sum_{C=1}^{C} y_c log(P_c) \tag{3}$$

where C is the number of classes, y_c is the true label (1 if class c is correct, 0 otherwise), and P_c is the predicted probability. This loss guides the model toward accurate predictions. The ability

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to effectively learn this mapping from context and input features to class probabilities, even with very few training examples provided in the context, is a key strength of TabPFN for few-shot classification.

Synthetic datasets for pre-training are generated using Structural Causal Models (SCMs) [33]. SCMs formalize causal relationships, with each variable X_i defined as:

$$X_i = f_i(pa(X_i), \epsilon_i) \tag{4}$$

where $pa(X_i)$ represents parent nodes (direct influencers), f_i is the causal function, and ϵ_i is noise, typically sampled from $N(0, \sigma^2)$.

As shown in Fig. 2, the core computational graph structure of TabPFN's synthetic data generation process for its pre-training phase is illustrated. This figure provides a visual representation of an example structural causal model (SCM) constructed based on predefined hyperparameters. The SCM encodes the computational function responsible for generating the synthetic dataset used in TabPFN's pre-training. Within this model: Nodes represent vectors that store intermediate data representations, Edges implement functions corresponding to specific connection types, enabling the flow and transformation of data throughout the graph. The process of generating the synthetic dataset unfolds in four distinct steps:

1. Initialization and Propagation: The root nodes of the SCM (the leftmost nodes in the figure) receive generated random noise variables as initial values. These values are then propagated forward through the depicted computational graph, with the functions along the edges transforming the data as it flows between nodes.

2. Sampling of Feature and Target Nodes: From the nodes within the populated graph structure, positions for feature nodes (labeled F) and target nodes (labeled T) are randomly selected. These nodes correspond to the input features and classification labels, respectively, of the synthetic dataset.

3. Extraction of Data Representations: Intermediate data representations (vectors) are extracted directly from the graph at the locations of the selected feature (F) and target (T) nodes. This forms the raw core of the synthetic dataset sample.

4. Post-processing: After being extracted from the graph in the previous step, the data vectors subsequently undergo post-processing. This typically involves standardization, quantization, warping, or other transformations to ensure the final dataset sample is suitable for the training process.

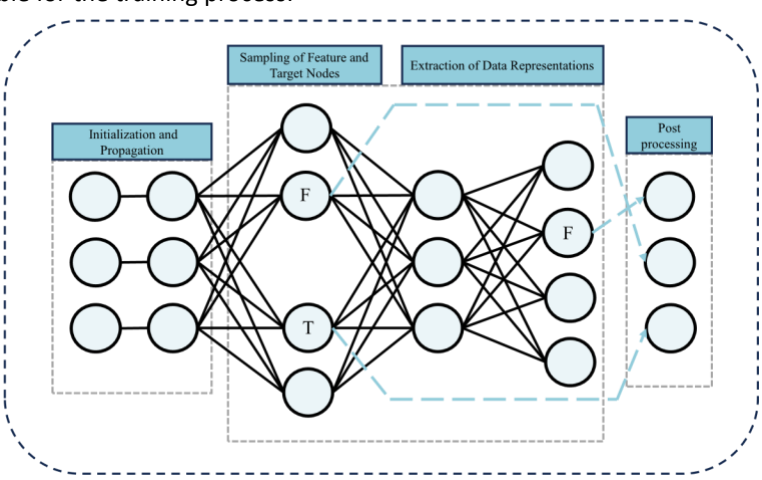

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Fig. 2. TabPFN computational graph and structure representation.

This structured methodology for generating diverse synthetic datasets lays a solid foundation for TabPFN's pre-training phase. The integration of Transformer attention, the approximation of Bayesian prediction, the cross-entropy loss minimization during pre-training, and the SCM-based data generation ultimately culminate in a robust tabular classification model. Crucially, this pre-trained model can then be applied to new few-shot classification tasks, like the Φ-OTDR event recognition in this study, making accurate predictions via ICL with only a small number of contextual examples from the target task.

The core of the proposed method is the application of TabPFN's ICL for few-shot classification. The complete workflow is schematically illustrated in Fig. 3.

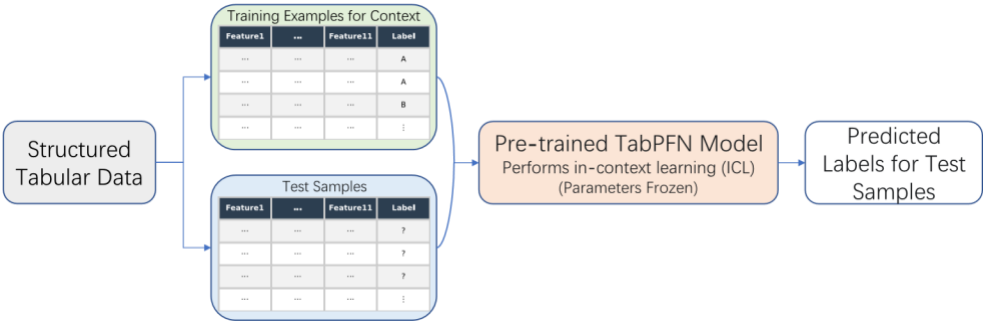

Fig. 3. Schematic of the ICL workflow for the proposed TabPFN-based few-shot classification method.

As depicted in Fig. 3, the ICL workflow starts with feature-engineered tabular data, which is logically partitioned into a small, labeled context set and an unlabeled test set. Both are provided as simultaneous input to the Pre-trained TabPFN Model. With its parameters frozen, the model leverages the context to directly infer the labels for the test samples in a single, inference-based forward pass, eliminating the need for task-specific training.

This in-context learning mechanism is the key to understanding the model's behavior: although the model's weights are fixed, its predictions are conditioned on the provided context set. Therefore, a larger and more informative context provides more evidence for the model to reason from during inference, leading to a more accurate prediction.

3. Data acquisition and classification experiments

3.1. Data source

The Φ-OTDR data for this study is sourced from the public dataset provided by the IoT Sensor Lab at Beijing Jiaotong University [13], which includes six event types (background, digging, knocking, water flow, shaking, and walking). Each event originally forms a 10,000 × 12 matrix (10,000 time-points, 12 spatial points). For our one-dimensional feature extraction approach, we selected the waveform from the 6th spatial point, typically near the event's center, to capture a strong vibrational signature. This selected raw waveform, consisting of 10,000 time-domain samples representing a 1-second duration with an effective 10 kHz sampling rate, was then used for feature extraction. This focus on a single spatial channel per event significantly reduces input dimensionality and subsequent computational load.

3.2. Feature extraction

A set of eleven time-domain and frequency-domain features, detailed in Table 1, were extracted from each Φ-OTDR waveform to construct a feature vector for classification. This

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tabular representation is designed to capture discriminative signal characteristics. By transforming the raw waveforms into this fixed-dimensional feature space, the proposed method becomes inherently adaptable to signals of varying original time durations or sampling rates, provided that these 11 features can be robustly computed.

Eleven descriptors are concatenated into a feature vector for each waveform, creating a structured tabular dataset feeding into our classification models. The overall data processing pipeline, encompassing spatial channel selection, feature extraction, and tabular data assembly, is depicted in Fig. 4.

Table 1. Extracted Features and Their Formulations[a]

Category	Feature Name	Description and formulation
Time-domain	Mean	Average amplitude of the signal. Formulation: $\frac{1}{N}\sum_{k=1}^{N} x_k$
	Standard Deviation (std)	Standard deviation of signal. Formulation: $\sqrt{\frac{1}{N-1}\sum_{k=1}^{N}(x_k - Mean)^2}$
	PeakToPeak	Peak-to-peak value. Formulation: $max(x) - min(x)$
	Skewness	Measure of signal asymmetry. Formulation: $\frac{\frac{1}{N}\sum_{k=1}^{N}(x_k - Mean)^3}{\sigma_N{}^3}$
	Kurtosis	Sharpness or flatness of the distribution. Formulation: $\frac{\frac{1}{N}\sum_{k=1}^{N}(x_k - Mean)^4}{\sigma_N{}^4}$
	RMS	Root mean square amplitude. Formulation: $\sqrt{\frac{1}{N}\sum_{k=1}^{N} x_k{}^2}$
	CrestFactor	Ratio of peak amplitude to RMS. Formulation: $\frac{max(\lvert x\rvert)}{RMS}$
	AutocorrLag1	Autocorrelation coefficient at lag 1. Formulation: $\frac{\sum_{k=2}^{N}(x_k - Mean)(x_{k-1} - Mean)}{\sum_{k=1}^{N}(x_k - Mean)^2}$
Frequency-domain	Spectral Centroid (sc)	Frequency "center of mass". Formulation: $\frac{\sum_{i=1}^{M} f_i \cdot A(f_i)}{\sum_{i=1}^{M} A(f_i)}$
	Spectral Width	Spectral bandwidth. Formulation: $\sqrt{\frac{\sum_{i=1}^{M}(f_i - sc)^2 \cdot A(f_i)}{\sum_{i=1}^{M} A(f_i)}}$
	Energy Spectrum	Average energy from spectrogram. Formulation: $\frac{1}{N_T}\sum_{t=1}^{N_T}\sum_{i=1}^{N_F}\lvert S(f,t)\rvert^2$

[a]NOTE: N is Number of time-domain samples in each waveform, k is the time-domain sample index, x_k is the amplitude of the k-th time-domain sample. M is Number of frequency bins. f is the frequency-bin index. $A(f_i)$ is its amplitude spectrum magnitude, N_T is the number of time frames, N_F is the number of frequency bins per frame, and $S(f,t)$ is the spectrogram coefficient. AutocorrLag1 is the normalized sample autocorrelation at lag 1 from MATLAB's autocorr function. σ_N is the population-like standard deviation of x, calculated as $\sqrt{\frac{1}{N}\sum_{k=1}^{N}(x_k - Mean)^2}$

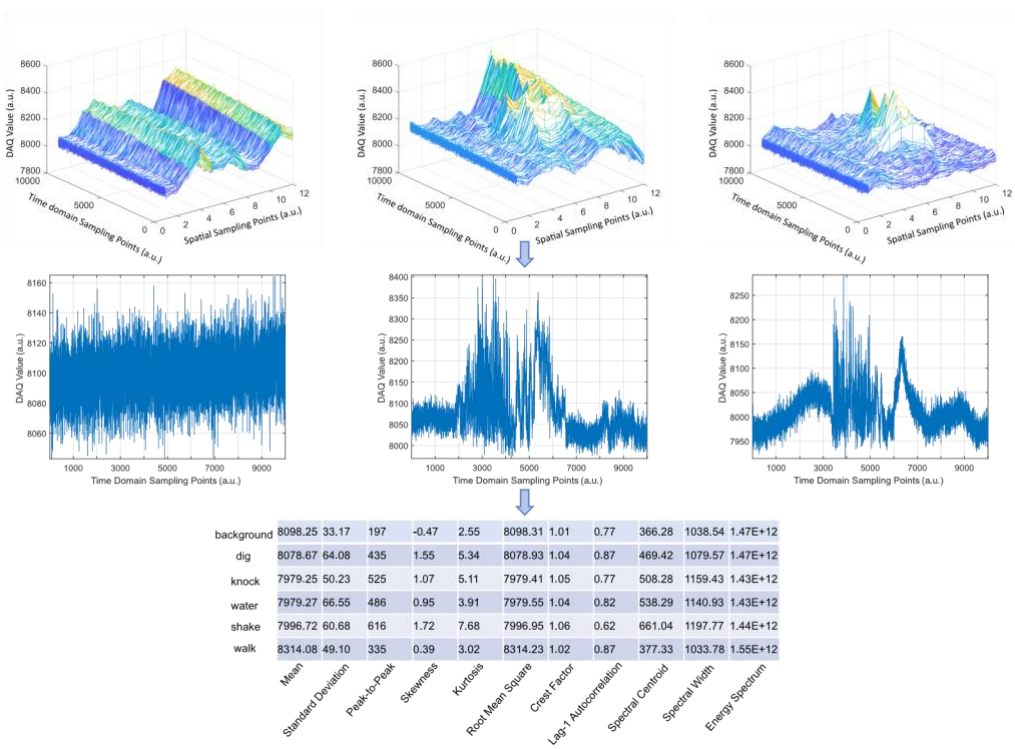

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Fig. 4. Data processing flowchart. Step 1: Selection of the 6th spatial sampling point. Step 2: Extraction of the eleven features (defined in Table 1) and tabularization of 1D waveforms.
For illustrative clarity, the flowchart depicts this process for three representative event types (e.g., Background, Digging, and Knocking) out of the six event types studied.

To quantitatively evaluate model performance in subsequent experiments, we define three standard metrics: Accuracy (ACC), macro-averaged F1-Score, and Area Under the Receiver Operating Characteristic Curve (AUC).

First, Accuracy (ACC) measures the proportion of correctly classified samples. It is defined as:

$$ACC = \frac{\textit{Number of Correct Predictions}}{\textit{Total Number of Predictions}} \tag{5}$$

Second, the F1-Score (Macro) is the unweighted average of the per-class F1-scores, making it robust to potential class imbalance. The F1-score is the harmonic mean of precision and recall:

$$F1 - \text{Score} = 2 \times \frac{\textit{Precision} \times \textit{Recall}}{\textit{Precision} + \textit{Recall}} \tag{6}$$

Finally, the AUC (OvR) is the Area Under the Curve, calculated using a One-vs-Rest (OvR) strategy for the multi-class problem. It reflects a model's ability to distinguish between classes, averaged across all classes.

3.3. *Five-fold cross-validation and independent test set validation*

To mitigate potential bias in few-shot learning, we employed a five-fold framework, evaluating the model with training sample sizes of 10 to 60 per class. Additionally, an independent test set validated the cross-validation results and assessed generalization to unseen data. Figure 5 shows the average accuracy (a) and ROC AUC (b) from the five-fold cross-validation.

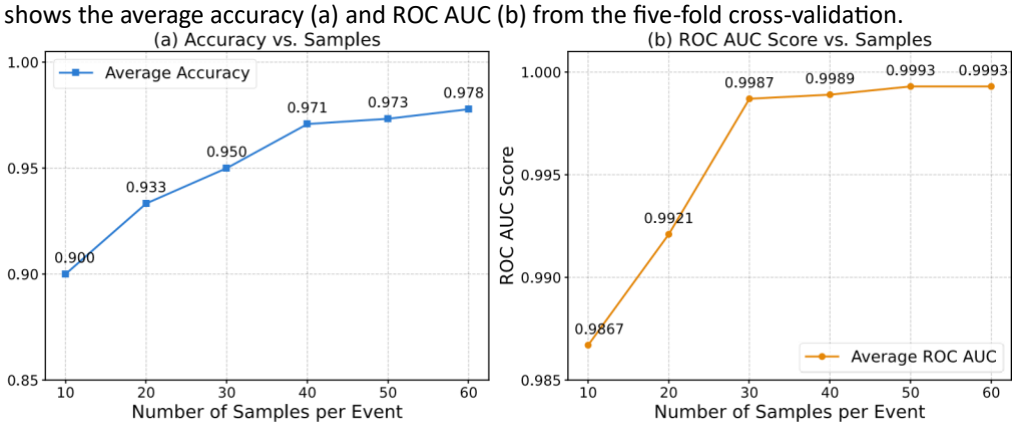

Fig. 5. Average Accuracy of five-fold cross-validation (a) and Average ROC AUC of five-fold cross-validation (b)

The results indicate model performance stabilizes as the sample size increases, confirming the robustness of the proposed method in few-shot scenarios.

The model's generalization was further validated on an independent test set, with results presented in Fig. 6. The performance trends for both accuracy and ROC AUC closely mirror those from the cross-validation, confirming the model's ability to effectively learn from limited data and generalize to new, unseen examples.

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Figure 7 presents the confusion matrices for TabPFN on the independent test set, illustrating its classification behavior at representative few-shot sample sizes.

In summary, through a dual validation approach employing both five-fold cross-validation and an independent test set, TabPFN consistently demonstrated strong learning capabilities and robust generalization in few-shot Φ-OTDR event classification. The results clearly show that TabPFN's overall classification performance generally improves and stabilizes as the number of

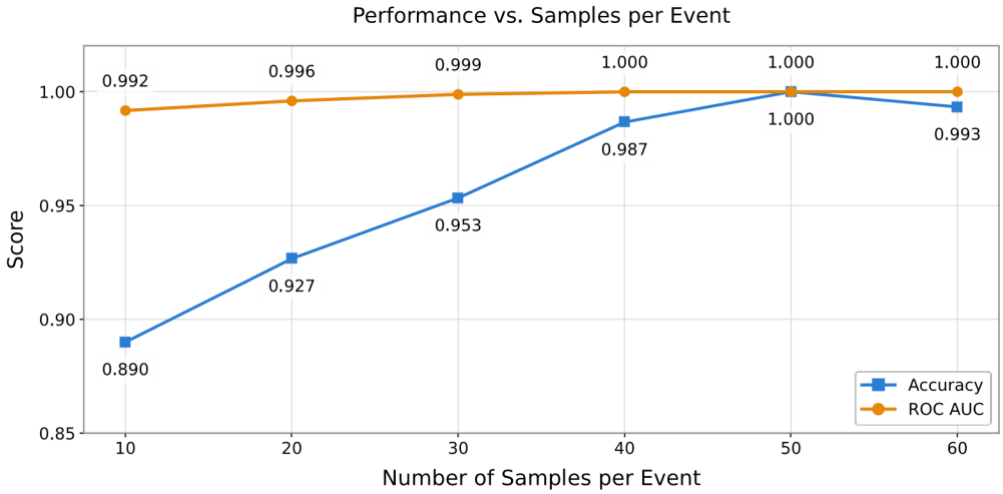

Fig. 6. Independent test set validation.

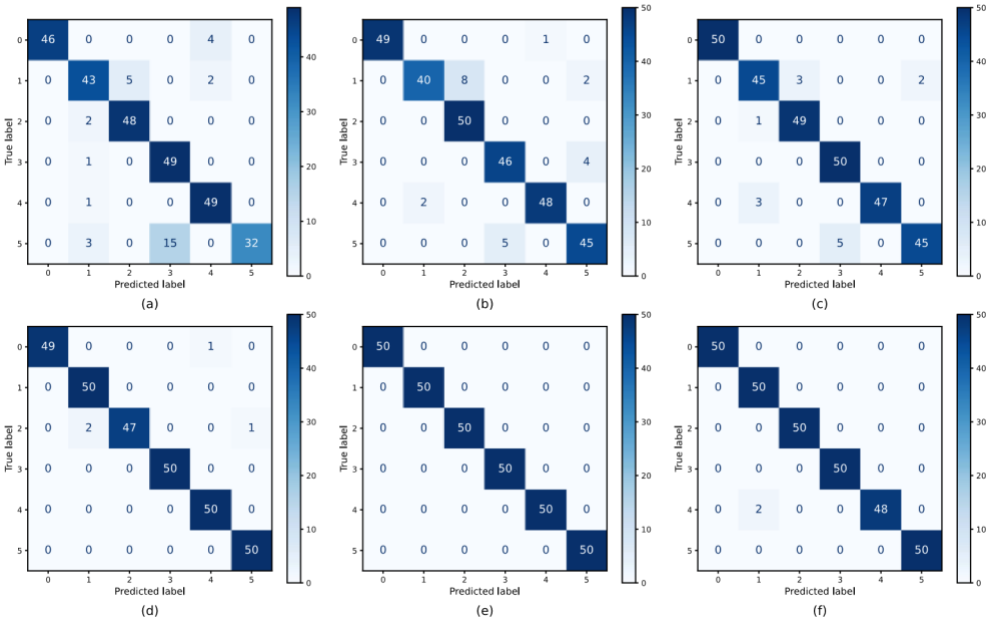

Fig. 7. Confusion Matrices at Representative Sample Sizes: (a) 10 samples/class, (b) 20 samples/class, (c) 30 samples/class, (d) 40 samples/class, (e) 50 samples/class, and (f) 60 samples/class

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context samples per class increases within the tested few-shot range (10 to 60 samples/class), reaching near-perfect classification on the independent test set with an optimal context size around 50 samples per class. While minor variations were observed between cross-validation and independent test set results, particularly at very low sample counts, the overall findings affirm TabPFN's effectiveness in leveraging minimal contextual information for accurate event recognition on unseen data.

3.4. Ablation experiments

We conducted ablation experiments to systematically evaluate the contribution of each of the 11 features. Within the same FSL and five-fold cross-validation framework, we iteratively removed one feature at a time and recorded the model's performance. Figure 8 illustrates the resulting changes in average accuracy and ROC AUC across different sample sizes.

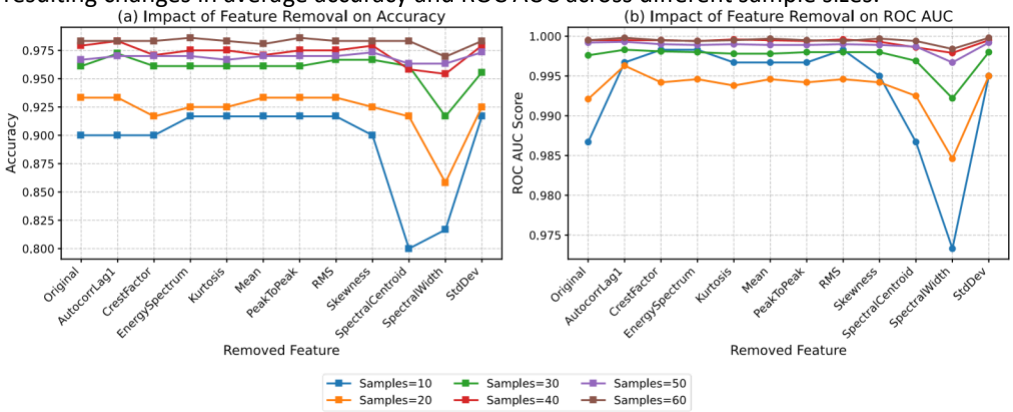

Fig. 8. Average accuracy of ablation tests (a) and Average ROC AUC of ablation tests (b).

The ablation study reveals a key insight: under few-shot conditions, removing certain features sometimes improved performance. This counterintuitive result suggests that a smaller, more targeted feature set can mitigate over-parameterization and better match the limited data scale. It highlights that feature quality can be more critical than quantity, and the optimal feature set may depend on the data scale and domain.

Notably, the removal of Spectral Width consistently led to a significant reduction in classification accuracy across most sample sizes (as seen in Fig. 8). This underscores its critical role in capturing the intensity and spread of phase disturbances within the Φ-OTDR signal. Similarly, Spectral Centroid, which reflects the dominant frequency distribution of the disturbance, also proved to be an important feature when removed. The physical interpretability of these individual spectral features aligns well with Φ-OTDR signal physics: the spectral centroid effectively captures the dominant frequency shift induced by an event, while the spectral width quantifies the multi-scale vibration bandwidth. The observed significant impact of removing either feature individually validates their importance in our feature set.

Furthermore, it is important to consider the interrelation between Spectral Centroid and Spectral Width. As defined in Table 1, the calculation of Spectral Width explicitly incorporates the Spectral Centroid, and both features describe aspects of the spectral energy distribution. This inherent relationship suggests that a degree of collinearity may exist between them for certain signal patterns. Consequently, the performance drop observed upon removing one of these features (e.g., Spectral Width) might partially reflect the loss of information that is also captured, to some extent, by the other (e.g., Spectral Centroid). This consideration adds a layer of nuance to the interpretation of individual feature impacts in the ablation study, highlighting their collective contribution to characterizing the signal spectrum.

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4. Comparative experiments

4.1. Experimental setup

To comprehensively evaluate our proposed TabPFN-based method, we compared it against two distinct categories of baseline models. The first category consists of three strong traditional machine learning models: XGBoost [34], LightGBM [35], and Random Forest (RF) [36]. The second category includes two representative meta-learning models adapted for this study: the classic Prototypical Network (ProtoNet) [37] and an attention-enhanced variant, SE-ProtoNet. The design of SE-ProtoNet was inspired by recent work applying attention mechanisms to Prototypical Networks for image-based Φ-OTDR data [19], and its core concept was adapted to the tabular framework used in this work by integrating a Squeeze-and-Excitation (SE) block. All experiments were conducted on the public Φ-OTDR dataset from the IoT Sensor Lab [13], using a unified feature extraction process (as described in Section 3) to ensure a fair comparison. The computational environment was a laptop equipped with an NVIDIA GeForce RTX 3070 Ti GPU, running Python 3.12.

To simulate varying degrees of data scarcity, we evaluated all models under few-shot conditions with training sets containing 10, 20, 30, 40, 50, and 60 samples per class. Model robustness was assessed using a five-fold cross-validation scheme for each sample size of the traditional baselines.

Furthermore, to evaluate the final generalization ability of the models, we pre-partitioned an independent test set comprising 300 samples (six classes, 50 samples per class) from the original dataset. This independent test set remained completely separate during all model training, fitting, and any parameter adjustment processes. For each model trained or fitted on a specific few-shot scale, we performed a single final performance evaluation on this independent test set.

For the traditional baseline models, key hyperparameters were manually tuned based on cross-validation performance, with others at their library defaults. The final parameters were set as follows:

XGBoost: max_depth = 6, learning_rate = 0.3, n_estimators = 100, objective='multi:softprob'.

LightGBM: boosting_type='gbdt', num_leaves = 31, learning_rate = 0.1, n_estimators = 100, min_child_samples = 20, random_state = 42, verbose = -1.

Random Forest (RF): n_estimators = 100, max_features='sqrt', random_state = 42.

For meta-learning baselines (ProtoNet and SE-ProtoNet), their final performance was assessed directly on the independent test set via a meta-testing procedure. Due to their episodic training nature, conventional five-fold cross-validation results are not reported for them. Key implementation details for these models are:

Embedding Network: A Multi-Layer Perceptron (MLP) with one hidden layer (32 units, ReLU activation, Batch Normalization) was used to map the 11-dimensional feature vectors into a 32-dimensional metric space.

Meta-Training: Models were trained for 100 epochs using the Adam optimizer with a learning rate of 5e-5. Each epoch consisted of 100 episodes, with each episode structured as a 6-way, 5-shot task (with 5 support samples and 3 query samples per class).

Meta-Testing: Final performance was averaged over 600 randomly generated test episodes from the independent test set, each with 5 support samples and 45 query samples per class.

The TabPFN model directly used its pre-trained weights for ICL fitting, without requiring additional hyperparameter tuning for this task.

We evaluated performance using three primary metrics: mean Accuracy (ACC), mean F1-Score (macro-averaged), and mean Area Under the ROC Curve (AUC). For efficiency, we also recorded the "Fit Time" and "Inference Time". Cross-validation results are reported as mean ± standard deviation.

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4.2. Cross-validation performance comparison

The detailed results of our five-fold cross-validation, including performance metrics and efficiency, are presented in Table 2. The trends for mean accuracy and F1-Score are visualized in Fig. 9.

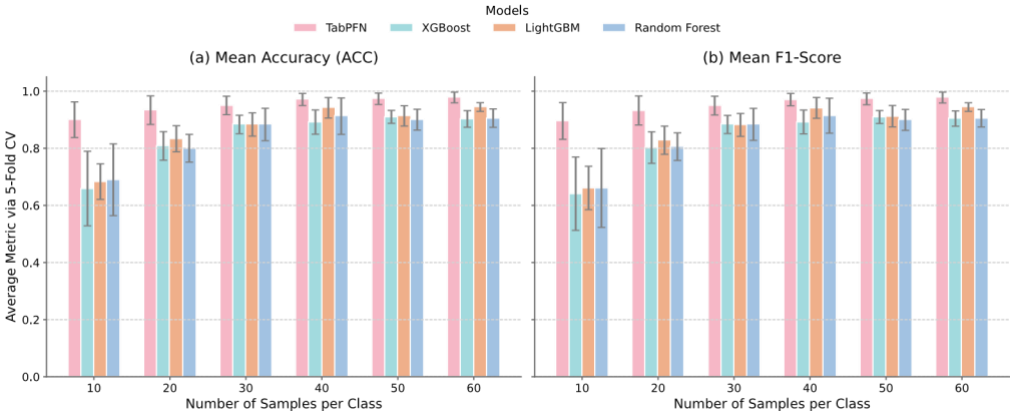

Fig. 9. Comparison of (a) mean accuracy (ACC) and (b) mean F1-Score (macro-averaged) for different models, obtained via five-fold cross-validation as a function of the number of training samples per class. Error bars represent ±1 standard deviation. Detailed results are presented in Table 2.

As shown in Table 2 and Fig. 9 within the cross-validation framework, the performance of all models generally improves as the number of training samples per class increases. TabPFN demonstrates strong learning capabilities across all tested few-shot scales. Particularly when the sample size is extremely small (e.g., 10 samples per class), TabPFN's mean accuracy (90.00% ± 6.24%), AUC (98.67% ± 1.87%), and F1-Score (89.56% ± 6.42%) are all significantly higher than the corresponding metrics for XGBoost, LightGBM, and Random Forest.

Notably, XGBoost and Random Forest exhibit larger standard deviations at 10 samples per class (e.g., accuracy standard deviations of 13.06% and 12.57%, respectively), indicating poorer stability under extreme few-shot conditions. As the sample size increases, the performance of all baseline models improves significantly. For instance, at 60 samples per class, LightGBM achieves an accuracy of 94.44% (±1.52%), while TabPFN reaches an accuracy of 97.78% (±1.88%) at this point, continuing to demonstrate its performance advantage in the few-shot regime. The trend of the F1-Score is largely consistent with that of accuracy, further confirming TabPFN's capability in balancing precision and recall.

In terms of efficiency, TabPFN was exceptionally fast to "fit," with a mean adaptation time of approximately 0.1 seconds, outperforming all baselines. For inference, however, the tree-based models (XGBoost, LightGBM, RF) were significantly faster, operating at the millisecond level, while the Transformer-based TabPFN required around 0.5 seconds. This trade-off highlights TabPFN's strength in rapid, on-the-fly adaptation versus the deployment efficiency of conventional tree models.

4.3. Independent test set performance comparison and generalization analysis

The final generalization performance of all models—TabPFN, traditional baselines, and metalearning baselines—was assessed on the held-out independent test set. The comprehensive

Table 2. Performance Comparison of Models using Five-Fold Cross-Validation across Varying Few-Shot Sample Sizes (Mean±Standard Deviation)

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Model	Samples per Class	Mean Accuracy (%)	Mean AUC (%)	Mean F1-Score (%)	Fit Time (s)	Inference Time (s)
TabPFN	10	90.00±6.24	98.67±1.87	89.56±6.42	0.0978	0.5402
	20	93.33±5.00	99.21±1.03	93.23±0.508	0.0901	0.5260
	30	95.00±3.24	99.87±0.16	94.94±0.330	0.0967	0.5403
	40	97.08±2.12	99.89±0.16	97.05±2.17	0.0855	0.4914
	50	97.33±2.00	99.93±0.07	97.29±2.05	0.0890	0.4796
	60	97.78±1.88	99.93±0.10	97.75±1.91	0.0868	0.4763
XGBoost	10	65.90±13.06	81.96±7.99	64.11±12.84	0.0575	0.0017
	20	80.83±5.00	97.10±1.37	80.25±5.52	0.0670	0.0016
	30	88.33±3.24	98.34±1.03	88.34±3.18	0.0684	0.0013
	40	89.17±4.25	98.90±1.39	89.21±4.18	0.0840	0.0012
	50	91.00±2.26	99.10±1.02	90.95±2.27	0.0944	0.0012
	60	90.28±2.91	98.73±0.69	90.40±2.67	0.1131	0.0013
LightGBM	10	68.33±6.24	93.00±4.61	66.11±7.59	0.2859	0.0022
	20	83.33±4.56	97.50±1.20	82.82±4.92	0.2900	0.0020
	30	88.33±4.08	98.22±1.09	88.17±4.01	0.2979	0.0020
	40	94.17±3.58	98.96±1.09	94.14±3.65	0.3077	0.0022
	50	91.33±3.56	99.03±0.60	91.23±3.76	0.3125	0.0022
	60	94.44±1.52	99.41±0.71	94.42±1.50	0.3274	0.0024
Random Forest	10	68.97±12.57	84.66±7.28	66.10±13.80	0.0914	0.0053
	20	80.00±4.86	96.74±2.49	80.58±4.83	0.0989	0.0061
	30	88.33±5.67	98.85±0.88	88.38±5.60	0.0936	0.0055
	40	91.25±6.3	98.77±2.19	91.41±6.11	0.0985	0.0049
	50	90.00±3.65	99.03±0.49	89.94±3.66	0.0996	0.0051
	60	90.56±3.22	98.69±0.49	90.51±3.07	0.1069	0.0052

results are summarized in Table 3. For clearer visualization, Fig. 10 compares TabPFN against the traditional baselines, while Fig. 11 benchmarks TabPFN against the meta-learning baselines.

Results from the independent test set (Table 3) confirm TabPFN's superior generalization over all baselines across few-shot scales, especially with minimal data. With only 10 samples per class, TabPFN achieved 89.0% accuracy and 99.17% AUC, while the best-performing baseline, Random Forest, only reached 69.33% accuracy and 92.10% AUC. This highlights TabPFN's remarkable capacity to leverage its pre-trained knowledge for accurate generalization from extremely sparse data.

As sample size increased, all models improved, but TabPFN reached perfect classification (100% accuracy) with just 50 samples per class. In contrast, the baseline models never matched this level of performance. For instance, the best-performing baseline, LightGBM, achieved its peak accuracy of 93.67% with 60 training samples, which was still substantially lower than

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TabPFN's. This persistent performance gap underscores the fundamental advantage of TabPFN's architecture for few-shot learning tasks.

The comparison with meta-learning baselines, presented in Fig. 11, provides further crucial insights. First, as dedicated FSL methods, both ProtoNet (84.04% accuracy at 10 samples/class) and SE-ProtoNet (83.87%) significantly outperform the traditional baselines (e.g., LightGBM at

Table 3. Performance Comparison of Models on the Independent Test Set across Varying Few-Shot Training/Fitting Sample Sizes

Model	Samples per Class	Accuracy (%)	AUC (%)	F1-Score (%)
TabPFN	10	89.00	99.17	88.73
	20	92.67	99.60	92.62
	30	95.33	99.89	95.32
	40	98.67	99.997	98.66
	50	100.0	100.0	100.0
	60	99.33	100.0	99.33
XGBoost	10	68.67	89.20	67.23
	20	71.33	86.51	70.84
	30	74.67	95.77	74.65
	40	83.67	97.01	83.37
	50	89.00	98.58	89.00
	60	91.33	99.11	91.37
LightGBM	10	67.67	92.83	67.23
	20	71.33	90.49	70.84
	30	82.33	97.59	82.89
	40	86.67	98.28	86.42
	50	89.33	98.74	89.45
	60	93.67	99.52	93.69
Random Forest	10	69.33	92.10	67.61
	20	71.67	92.84	71.17
	30	76.67	96.98	76.66
	40	81.33	96.51	81.08
	50	85.67	98.05	85.56
	60	89.33	98.45	89.39
ProtoNet	10	84.04	98.35	84.01
	20	86.37	98.30	86.30
	30	92.99	99.52	92.96
	40	94.18	99.66	94.14

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	50	97.40	99.89	97.40
	60	98.25	99.98	98.25
SE-ProtoNet	10	83.87	98.19	83.68
	20	84.53	96.35	84.38
	30	93.28	99.52	93.29
	40	94.55	99.76	94.49
	50	97.25	99.88	97.26
	60	97.40	99.94	97.40

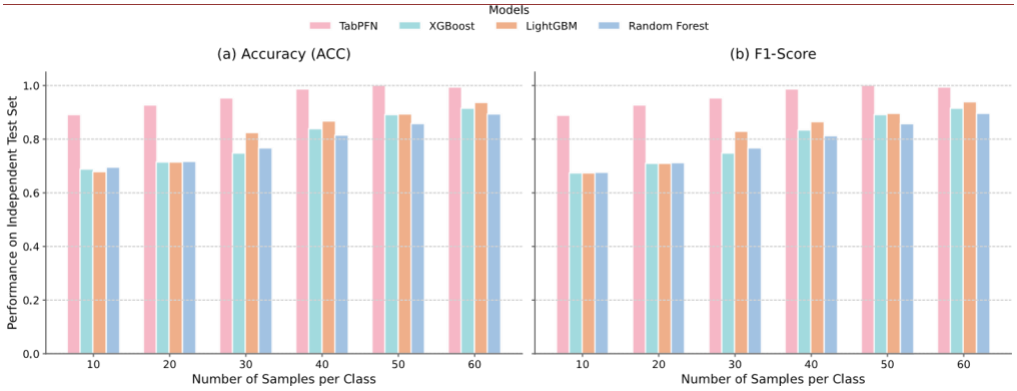

Fig. 10. Comparison of (a) Accuracy (ACC) and (b) F1-Score (macro-averaged) for TabPFN and traditional baselines on the independent test set, evaluated with varying training samples per class. Detailed results are in Table 3.

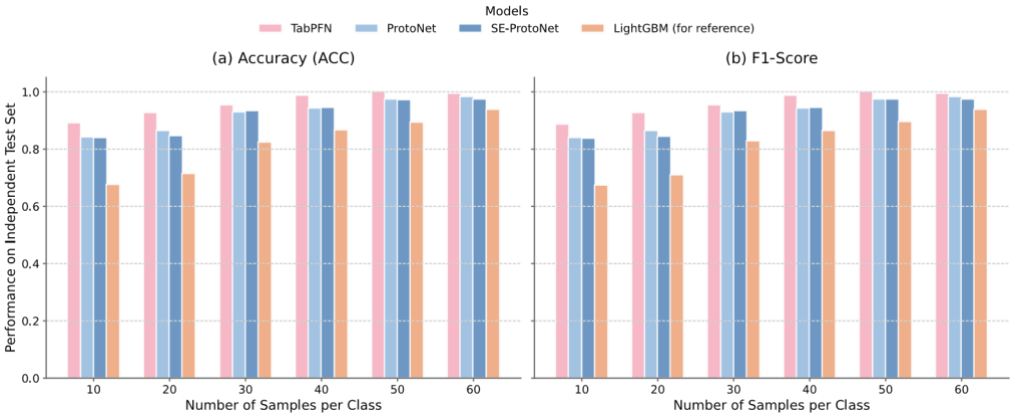

Fig. 11. Comparison of (a) Accuracy (ACC) and (b) F1-Score (macro-averaged) for TabPFN and meta-learning baselines on the independent test set, evaluated with varying training samples per class. Results are in Table 3.

67.67%), confirming the effectiveness of the meta-learning paradigm for this task. However, the addition of the SE attention mechanism in SE-ProtoNet did not yield a performance improvement over the standard ProtoNet; its accuracy was slightly lower at most sample points. This suggests that for this low-dimensional feature space, the extra complexity may not be beneficial.

Most importantly, the proposed TabPFN method consistently outperforms both the traditional and the meta-learning baselines. For instance, at 10 samples per class, TabPFN's accuracy (89.00%) is substantially higher than that of the standard Prototypical Network (84.04%). This result strongly suggests that for this task, TabPFN's paradigm of "large-scale general pre-training + ICL" is more effective than the "meta-learning from scratch on in-domain data" paradigm of Prototypical Networks. TabPFN's powerful prior knowledge provides a more data-efficient starting point for few-shot adaptation.

When the number of context samples exceeded 50 per class (e.g., 60 samples), TabPFN's accuracy on the independent test set slightly fluctuated (99.33%), below the perfect score at 50 samples (though AUC remained 1.0). This fluctuation did not appear in cross-validation but emerged on the independent set, suggesting that TabPFN's ICL mechanism may have an optimal contextual range for this task.

5. Conclusion

This paper introduced and validated a novel Φ-OTDR event classification method by integrating a few-shot learning (FSL) approach with the pre-trained TabPFN model, effectively leveraging tabular machine learning. Addressing challenges of data scarcity and computational cost in Φ-OTDR monitoring, the proposed method demonstrated outstanding performance under small-sample conditions. Its rapid adaptation, with fitting completed in seconds, highlights its potential for real-time applications and edge deployment. Rigorous validation using fivefold cross-validation and an independent test set confirmed its effectiveness. Crucially, when benchmarked against both traditional machine learning models and representative meta-learning FSL baselines, our method consistently demonstrated superior performance, particularly in extreme few-shot scenarios. Notably, even with as few as 10 samples per class, the method achieved a mean cross-validation accuracy of 90.0% ± 6.24% and an independent accuracy of 89.0%, reaching 100% with 50 samples per class. This key result suggests that the "large-scale general pre-training + ICL" paradigm of TabPFN is a more powerful and data-efficient FSL strategy for this task than meta-learning from scratch. Ablation experiments also confirmed the critical role of frequency-domain features, particularly spectral width and centroid, aligning with the physical principles of Φ-OTDR sensing.

While this study demonstrates an effective solution, several areas warrant future investigation to enhance practical applicability and robustness: 1) exploring dynamic or learned feature extraction for more diverse event signatures and complex noise environments; 2) reducing TabPFN's inference time to the millisecond level for ultra-low latency applications; and 3) validating the scheme on broader datasets covering more event types, varying SNRs, and different sensing configurations. Addressing these challenges would further solidify TabPFN's applicability for real-world deployments.

Funding. National Natural Science Foundation of China (62275185); Jiangsu Province Innovation and Entrepreneurship Team Program (JSSCTD202343).

Disclosures. The authors declare no conflicts of interest related to this article.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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